

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 3 Lesson 1 Exercise 4: More with Relationships

Identify Hierarchical, Recursive and Arc Relationships (S3L1 Objective 5)

This exercise uses the additional scenario and meeting transcript that was provided in the previous lessons.

Part 1 - Recursive relationships

A recursive relationship allows you to represent a relationship that an entity has with itself.

A recursive relationship requires that one attribute in the entity be dependent on another attribute in the entity. Ensure that both attributes are present in the entity before you draw the recursive relationship.

Using the given scenario identify and draw any recursive relationships that you identified on your ERD.

Part 2 - Arc relationships

An arc is an exclusive relationship group, which is defined such that only one of the relationships can exist for any instance of an entity.

All relations included in an arc should belong to the same entity and should have the same cardinality.

In the previous exercise you created a supertype/subtype for customer that showed how a customer can be either an individual or team representative. One other way of showing this would have been using an arc.

As a separate ERD that only details the customer information show how customer could have been represented using an arc.